

Program 6:

Working with Gradle: Setting Up a Gradle Project, Understanding Build Scripts (Groovy and Kotlin DSL), Dependency Management and Task Automation

Step 1: create directory using command **mkdir MyGradleApp** and enter the directory using the command **cd MyGradleApp**

```
samiksha@samiksha:~/Devops6D$ mkdir MyGradleapp
samiksha@samiksha:~/Devops6D$ cd MyGradleapp
samiksha@samiksha:~/Devops6D/MyGradleapp$
```

Step 2: Type the command **gradle init --type java-application**

```
samiksha@samiksha:~/Devops6D/MyGradleapp$ gradle init --type java-application

Enter target Java version (min: 7, default: 21): 21

Project name (default: MyGradleapp): MyGradleapp

Select application structure:
  1: Single application project
  2: Application and library project
Enter selection (default: Single application project) [1..2] 1

Select build script DSL:
  1: Kotlin
  2: Groovy
Enter selection (default: Kotlin) [1..2] 2

Select test framework:
  1: JUnit 4
  2: TestNG
  3: Spock
  4: JUnit Jupiter
Enter selection (default: JUnit Jupiter) [1..4] 1

Generate build using new APIs and behavior (some features may change in the next minor release)? (default: no) [yes, no] no

> Task :init
To learn more about Gradle by exploring our Samples at https://docs.gradle.org/8.7/samples/sample_building_java_applications.html

BUILD SUCCESSFUL in 2m 9s
1 actionable task: 1 executed
<-----> 0% WAITING
> IDLE
samiksha@samiksha:~/Devops6D/MyGradleapp$
```

Step 3: Go to app directory to find build.gradle. Open build.gradle and type the code in build.gradle

```
task hello {
    doLast {
        println '6D Devops Class'
    }
}
```

Step 4: type the command **gradle build** to compile the source code and package the app.

```
samiksha@samiksha:~/Devops6D/MyGradleapp$ gradle build
Starting a Gradle Daemon, 1 incompatible and 1 stopped Daemons could not be reused, use --status for details
Path for java installation '/usr/lib/jvm/openjdk-11' (Common Linux Locations) does not contain a java executable
Path for java installation '/usr/lib/jvm/openjdk-17' (Common Linux Locations) does not contain a java executable

BUILD SUCCESSFUL in 1m 5s
7 actionable tasks: 7 executed
samiksha@samiksha:~/Devops6D/MyGradleapp$
```

Step 5: type the command **gradle run** to execute the default Hello World application

```
samiksha@samiksha:~/Devops6D/MyGradleapp$ gradle run
Path for java installation '/usr/lib/jvm/openjdk-17' (Common Linux Locations) does not contain a java executable
Path for java installation '/usr/lib/jvm/openjdk-11' (Common Linux Locations) does not contain a java executable

> Task :app:run
Hello World!

BUILD SUCCESSFUL in 1s
2 actionable tasks: 1 executed, 1 up-to-date
```

Step 6: type the command **gradle <task-name>** to run the task

```
samiksha@samiksha:~/Devops6D/MyGradleapp$ gradle hello

> Task :app:hello
6D Devops Class

BUILD SUCCESSFUL in 1s
1 actionable task: 1 executed
```

Program 7:

Practical Exercise: Build and Run a Java Application with Maven, Migrate the Same Application to Gradle

Step 1: To build a basic Java Application with Maven Project, run following command:
mvn archetype:generate -DgroupId=com.example -DartifactId=MyMaventoGradle -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
samiksha@samiksha:~/Devops6D$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMaventoGradle -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: /home/samiksha/Devops6D
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMaventoGradle
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/samiksha/Devops6D/MyMaventoGradle
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 6.666 s
[INFO] Finished at: 2026-03-25T11:57:45+05:30
[INFO] -----
```

Step 2: enter the MyMaventoGradle directory and open the pom.xml file.

```
samiksha@samiksha:~/Devops6D$ cd MyMaventoGradle
samiksha@samiksha:~/Devops6D/MyMaventoGradle$ ls
pom.xml  src
samiksha@samiksha:~/Devops6D/MyMaventoGradle$ gedit pom.xml
```

Add the code in the pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-
v4_0_0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.example</groupId>

<artifactId>MyMavenApp</artifactId>

<packaging>jar</packaging>

<version>1.0-SNAPSHOT</version>

<name>MyMavenApp</name>

<url>http://maven.apache.org</url>
```

```
<!-- Properties: Customize Java version or plugin versions -->
<properties>
<maven.compiler.source>11</maven.compiler.source>
<maven.compiler.target>11</maven.compiler.target>
</properties>
<dependencies>
  <dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>3.8.1</version>
    <scope>test</scope>
  </dependency>
</dependencies>
<!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
```

```

<version>2.22.2</version>

</plugin>

<plugin>

    <groupId>org.apache.maven.plugins</groupId>

    <artifactId>maven-jar-plugin</artifactId>

    <version>3.1.0</version>

    <configuration>

        <archive>

            <manifestEntries>

                <Main-Class>com.example.MainClass</Main-Class>

            </manifestEntries>

        </archive>

    </configuration>

</plugin>

</plugins>

</build>

</project>

```

Step 3 : Run **mvn clean install** to compile your project

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.051 s - in com.example.AppTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenApp ---
[INFO] Building jar: /home/samiksha/Devops6D/MyMaventoGradle/target/MyMavenApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenApp ---
[INFO] Installing /home/samiksha/Devops6D/MyMaventoGradle/target/MyMavenApp-1.0-SNAPSHOT.jar to /home/samiksha/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.jar
[INFO] Installing /home/samiksha/Devops6D/MyMaventoGradle/pom.xml to /home/samiksha/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.pom
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.933 s
[INFO] Finished at: 2026-03-25T12:07:52+05:30
[INFO] -----

```

Step 4: Type the command **mvn exec:java -Dexec.mainClass="com.exempls.App"** to execute

```

samiksha@samiksha:~/Devops6D/MyMaventoGradle$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenApp >-----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenApp ---
Hello World!
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.662 s
[INFO] Finished at: 2026-03-25T12:10:12+05:30
[INFO]
samiksha@samiksha:~/Devops6D/MyMaventoGradle$ █

```

Step 5: Type the command **gradle init --type pom** to initialize a Gradle Project using gradle init command with type as pom, command converts pom.xml structure to Gradle's build.gradle.

```

samiksha@samiksha:~/Devops6D/MyMaventoGradle$ gradle init --type pom

Select build script DSL:
  1: Kotlin
  2: Groovy
Enter selection (default: Kotlin) [1..2] 2

Generate build using new APIs and behavior (some features may change in the next minor release)? (default: no) [yes, no] no

> Task :init
Maven to Gradle conversion is an incubating feature.
For more information, please refer to https://docs.gradle.org/8.7/userguide/migrating_from_maven.html in the Gradle documentation.

BUILD SUCCESSFUL in 1m 54s
1 actionable task: 1 executed
samiksha@samiksha:~/Devops6D/MyMaventoGradle$ █

```

Type the ls command to see all the files and directories.

```

samiksha@samiksha:~/Devops6D/MyMaventoGradle$ ls
build.gradle  gradle  gradlew  gradlew.bat  pom.xml  settings.gradle  src  target
samiksha@samiksha:~/Devops6D/MyMaventoGradle$ █

```


